

Duc Toan Nguyen

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EDUCATION

- Aug 2025 - Present **Ph.D. in Electrical and Computer Engineering (ECE)**
Rice University, Houston, TX
George R. Brown School of Engineering and Computing
- May 2025 **Bachelor of Science in Mathematics**
Bachelor of Science in Computer Science
Texas Christian University (TCU), Fort Worth, TX
College of Science and Engineering, John V. Roach Honors College

RESEARCH EXPERIENCE

Rice University, Dept. of Electrical and Computer Engineering Aug 2025 - Present

- **Role:** Graduate Researcher (Distributed Fréchet Regression on the Bures-Wasserstein Manifold)
- **Advisor:** Dr. César A. Uribe
- Leverage the intrinsic Riemannian geometry of the **Bures-Wasserstein metric** to design efficient optimization algorithms for **network regression** on graph Laplacians.
- Derive rigorous **bottleneck conditions** characterizing stationary points of the **Fréchet mean** optimization problem with Wasserstein metric.
- Apply network regression methods to real-world datasets, specifically modeling **diffusion tensor MRI**, to advance computational methods in **neuroscience**.
- Propose the **first** intrinsic **decentralized Riemannian optimization** method establishing **exact convergence** in both network consensus and function error for convex functions on manifolds.

Independent Research (Collaboration with IBM/Lehigh) Apr 2024 - May 2025

- **Role:** Researcher (Variance-Reduced Shuffling Stochastic Optimization)
- **Advisors:** Dr. Lam M. Nguyen (IBM Research), Dr. Trang H. Tran (Lehigh University)
- Propose a novel variance-reduced optimization method by applying shuffling paradigms to SARAH, creating a protocol that is practically efficient for **large-scale machine learning** workloads.
- Prove a **faster convergence rate** for the proposed method compared to existing state-of-the-art algorithms, directly addressing **AI computational efficiency**.
- Develop an inexact version for Expectation Maximization (EM) problems with rigorous proofs of complexity comparable to leading stochastic methods.

Texas Christian University, Dept. of Computer Science Jan 2023 - May 2025

- **Role:** Research Assistant (GO2AI Project)
- **Advisors:** Dr. Liran Ma, Dr. Ze-li Dou
- Implement Monte Carlo Tree Search (MCTS) and Convolutional Neural Networks (CNNs) into the policy architecture of distributed **AI agents** (Go).
- Optimize **reinforcement learning** algorithms by enhancing the training pipeline through distributed learning techniques.
- Implement **Grad-CAM (Explainable AI/XAI)** to visualize and interpret black-box learning processes, evolving the transparency of AI decision-making.

- **Role:** Honors Research Scholar (Geodesic Nets - Construction and Existence)
- **Advisor:** Dr. Ken Richardson
- Develop a first-of-its-kind heuristic algorithm to construct approximate Steiner Trees for arbitrary point sets, reducing total network length by 2% compared to Minimum Spanning Trees.
- Apply the algorithm to optimize network architecture designs, with direct implications for efficient **communication infrastructures** and **genetic network analysis**.
- Prove theoretical conditions for the existence of Fermat points in triangles on general 2-D surfaces.

Rice University, Dept. of Statistics (REU)

May 2023 - July 2023

- **Role:** Undergraduate Researcher (STAT-DATASCI REU 2023)
- **Advisor:** Dr. Eric C. Chi
- Investigate the sensitivity of optimal tuning parameters to noise levels in Nonnegative Matrix Factorization (NMF), a critical technique for **dimensional reduction**.
- Spearhead the development of a novel algorithm, “Square-Root Min-Vol NMF,” and establishing its rigorous convergence guarantees.
- Validate the algorithm on large-scale datasets of **hyperspectral images**, demonstrating superior error reduction compared to recent methods, with applications in **biomedical imaging**.

ACADEMIC PUBLICATIONS

- **Nguyen, D. T.** & Uribe, C. A. “Fréchet Regression on the Bures-Wasserstein Manifold.” (*In Revision for the Conference on Neural Information Processing Systems (NeurIPS) 2026*).
- **Nguyen, D. T.** & Uribe, C. A. “Intrinsic Decentralized Stochastic Riemannian Optimization on Manifolds with Bounded Sectional Curvature.” (*Accepted for IEEE Control Systems Letters*).
- **Nguyen, D. T.**, Tran, T. H., & Nguyen, L. M. “Adjusted Shuffling SARAH: Advancing complexity analysis via dynamic gradient weighting.” (*Accepted for Journal of Optimization Theory and Applications - JOTA*).
- **Nguyen, D. T.**, & Chi, E. C. “Towards tuning-free minimum-volume nonnegative matrix factorization.” *Proceedings of the 2024 SIAM International Conference on Data Mining (SDM24)*.
- **Nguyen, D. T.** “On the existence of a balanced vertex in geodesic nets with three boundary vertices.” *Journal of Geometry*, 116.3 (2025): 36.
- **Nguyen, D. T.** “Geodesic Nets - Construction and Existence.” (*Outstanding Honors Thesis*).

HONORS & AWARDS

- 2024 **Top 300** (of 3,988 students) – 85th William Lowell Putnam Mathematical Competition
- 2026 **Pi Mu Epsilon Outstanding Speaker Award** – Joint Mathematics Meetings (JMM)
- 2025 **Senior Scholar** – Department of Mathematics, TCU
- 2025 **Best Undergraduate Poster** – TCU Student Research Symposium
- 2025 **Finalist**, Best Honors Thesis Presentation (Boller Competition) – TCU
- 2025 **Student Travel Funding** – Pi Mu Epsilon (PME) for JMM 2025
- 2024 **SIAM Student Travel Award** – SIAM Int. Conference on Data Mining (SDM24)
- 2023 **Outstanding Session Presentation** – Comp. Math & Operations Research, GCURS
- 2022 **Top 500** (of 3,415 students) – 83rd William Lowell Putnam Mathematical Competition
- 2025 Member, **Pi Mu Epsilon** (TCU Texas Alpha chapter)
- 2025 Member, **Upsilon Pi Epsilon** (TCU chapter)

PROFESSIONAL ACTIVITIES

- **Reviewer**, *The Conference on Neural Information Processing Systems (NeurIPS) 2026* - Top-tier conference in Machine Learning.
- **Reviewer**, *Journal of Machine Learning Research (JMLR)* - Top-tier journal in Machine Learning.
- **Reviewer**, *International Conference on Learning Representations (ICLR) 2026* - Top-tier conference in Machine Learning.
- **Reviewer**, *Journal of Optimization Theory and Applications (JOTA)* - Top-tier journal in Mathematical Optimization.

POSTERS & PRESENTATIONS

- **Nguyen, D. T.** “Geodesic Nets - Construction and Existence.” *PME Contributed Session on Research by Undergraduates*, JMM, Jan 2026.
- **Nguyen, D. T.** & Chi, E. C. “Towards Tuning-Free Minimum-Volume Nonnegative Matrix Factorization.” *AMS Contributed Papers Session, Numerical Analysis I*, JMM, Jan 2025.
- **Nguyen, D. T.** “On the existence of a balanced vertex in geodesic nets with three boundary vertices.” *AMS - PME Undergraduate Student Poster Session*, JMM, Jan 2025.
- **Nguyen, D. T.** & Chi, E. C. “Towards Tuning-Free Minimum-Volume Nonnegative Matrix Factorization.” *SIAM Conference on Mathematics of Data Science (MDS24)*, Oct 2024. (Poster).
- **Nguyen, D. T.** “Towards Tuning-Free Minimum-Volume Nonnegative Matrix Factorization.” *Summer school on Bayesian learning and network analysis*, VIASM, Hanoi, July 2024. (Poster).
- **Nguyen, D. T.** & Chi, E. C. “Towards Tuning-Free Minimum-Volume Nonnegative Matrix Factorization.” *SIAM International Conference on Data Mining (SDM24)*, Apr 2024. (Slides/Poster).
- **Nguyen, D. T.** “A Majorization-Minimization Variant For Minimum-Volume Nonnegative Matrix Factorization.” *National Collegiate Research Conference (NCRC)*, Harvard University, Jan 2024.
- **Nguyen, D. T.** “Towards Tuning-Free Minimum-Volume Nonnegative Matrix Factorization.” *Gulf Coast Undergraduate Research Symposium (GCURS)*, Rice University, Oct 2023.
- **Nguyen, D. T.** “Searching for networks of minimum length.” *Research and Creative Activities Week*, TCU, Sep 2023. (Poster).
- Leath, H., Good, B., Fahimi, S., **Nguyen, D. T.**, Ma, L., & Dou, Z. “The Sybil in AI: The Many Personalities of a Go Playing Model.” *Research and Creative Activities Week*, TCU, Sep 2023. (Poster).
- **Nguyen, D. T.** “Geodesic Nets construction using Genetic Algorithm.” *Student Research Symposium (SRS)*, TCU, April 2023. (Poster).

TECHNICAL SKILLS

Languages	Python, Java, MATLAB, R, MySQL, C, C++, HTML, JS, PHP
AI/ML	PyTorch, Scikit-learn, NetworkX, NumPy, Pandas, Captum, TensorFlow
Systems	Linux, MacOS, Windows, Distributed Computing